can occur by two methods: 1. Biological; 2. Non-Biological (Figure 12.5).

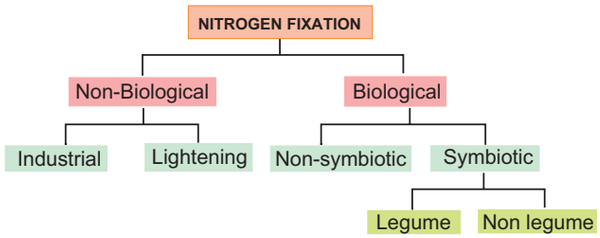


Figure 12.5: Nitrogen fixation

Activity

Preparation of Solution Culture to find out Mineral Deficiency

1. Take a glass jar or polythene bottle and cover with black paper (to prevent algal growth and roots reacting with light).
2. Add nutrient solution.
3. Fix a plant with the help of split cork.
4. Fix a tube for aeration.
5. Observe the growth by adding specific minerals.

12.7.1 Non – Biological nitrogen fixation

- Nitrogen fixation by chemical process in industry.
- Natural electrical discharge during lightening fixes atmospheric nitrogen.

12.7.2 Biological nitrogen fixation

Symbiotic bacterium like *Rhizobium* fixes atmospheric nitrogen. Cyanobacteria found in Lichens, *Anthoceros, Azolla* and coralloid roots of *Cycas* also fix nitrogen. Non-symbiotic (free living bacteria) like *Clostridium* also fix nitrogen.

a. Symbiotic nitrogen fixation

i. Nitrogen fixation with nodulation

Rhizobium bacterium is found in leguminous plants and fix atmospheric nitrogen. This kind of symbiotic association is beneficial for both the bacterium and plant. Root nodules are formed due to bacterial infection. *Rhizobium* enters into the host cell and proliferates, it remains separated from the host cytoplasm by a membrane (Figure 12.6).

Iron and Manganese toxicity

Iron and Manganese exhibit competitive behaviour. Deficiency of Fe and Mn shows similar symptoms. Iron toxicity will affect absorption of manganese. The possible reason for iron toxicity is excess usage of chelated iron in addition with increased acidity of soil (PH less than 5.8) Iron and manganese toxicity will be solved by using fertilizer with balanced ratio of Fe and Mn.

Stages of Root nodule formation:

1. Legume plants secretes phenolics which attracts *Rhizobium*.
2. *Rhizobium* reaches the rhizosphere and enters into the root hair, infects the root hair and leads to curling of root hairs.
3. Infection thread grows inwards and separates the infected tissue from normal tissue.



Figure 12.6: Rhizobium (Bacteroid) in root nodule

4. A membrane bound bacterium is formed inside the nodule and is called **bacteroid**.
5. Cytokinin from bacteria and auxin from host plant promotes cell division and leads to nodule formation

Activity

- Collect roots of legumes with root nodules.
- Take cross section of the root nodule.
- Observe under microscope. Discuss your observations with your teacher.

Non-Legume

Alnus and Casuarina contain the bacterium *Frankia. Psychotria* contains the bacterium *Klebsiella*.

ii. Nitrogen fixation without nodulation

The following plants and prokaryotes are involved in nitrogen fixation.

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